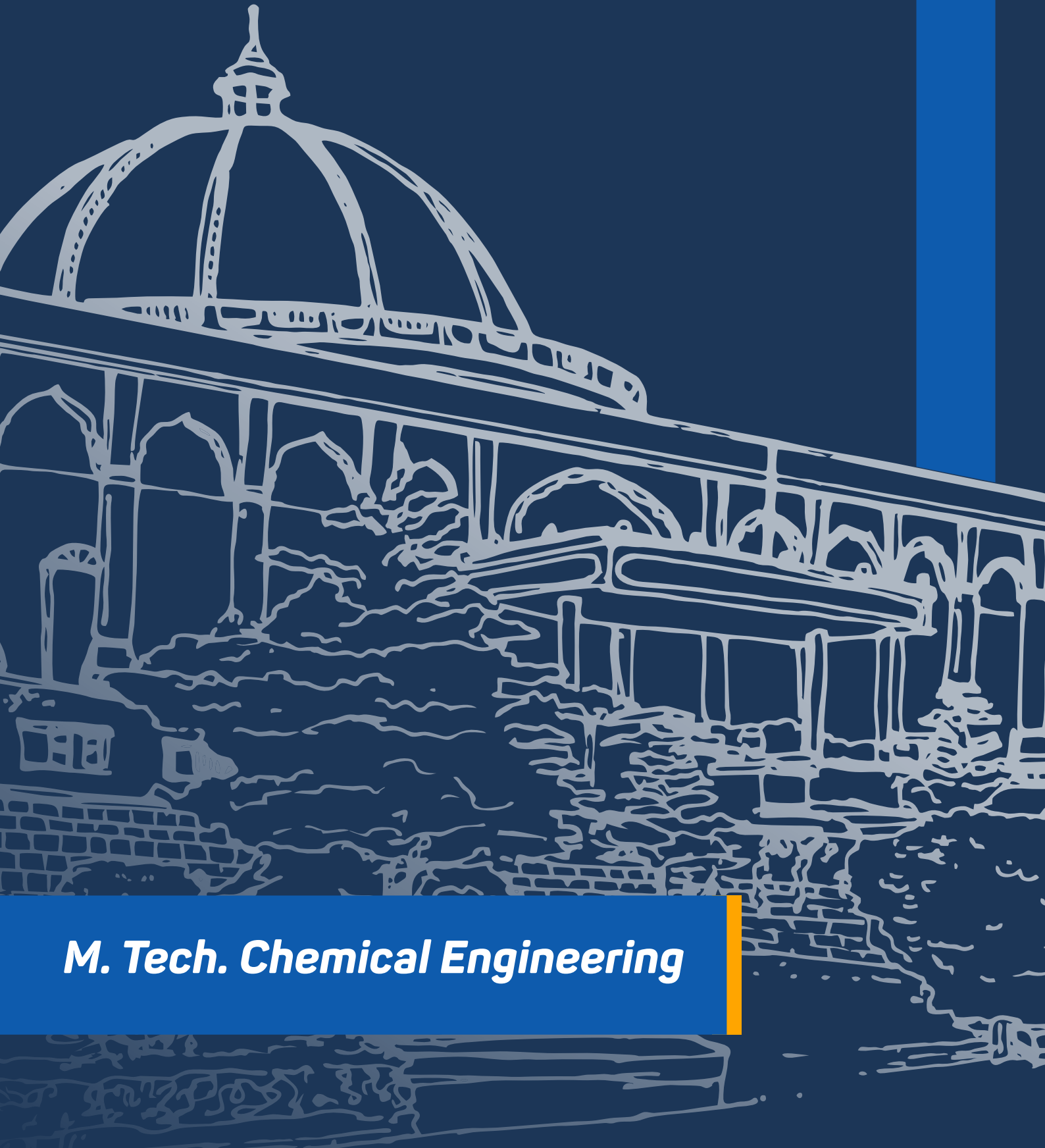




M. Tech. Chemical Engineering

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	DEPARTMENT OF CHEMICAL ENGINEERING
Program Name	M. Tech. Chemical Engineering

+++++ COURSE BASKET +++++

Course Type	Description
Programme Core [PC]	Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU
Programme Electives [PE]	Open electives under the programme allow students to specialise in a particular area connected to their major.
University Core [UC]	Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students.
University Electives [UE]	Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core.

Semester I

Semester	Course Type	Course Name / Course Title	Total Credits
I	PM	Chemical Process Synthesis and Design	4
I	PM	Numerical Methods and Computational Techniques	4
I	PM	Renewables and Energy Systems	4
I	PM	Research Methodology for Chemical Engineers	4
I	UC	Scientific Studies of Mind, Matter, Spirit and Consciousness	2
I	PM	Computer Aided Chemical Engineering-I (CACH-E-I)	2
I	UC	Yoga	1
			21

Semester II

Semester	Course Type	Course Name / Course Title	Total Credits
II	PM	Advanced Chemical Reaction Engineering	4
II	PM	Process Control and Automation	4
II	PM	Process Systems Engineering	4
II	PE	Professional Elective-I	4
II	UC	Peace Building: Global Initiatives	2
II	PM	Computer Aided Chemical Engineering-II (CACH-E-II)	1
II	PR	Seminar	2
		Total	21

Semester - III

Semester	Course Type	Course Name / Course Title	Total Credits
III	PE	Professional Elective-II®	4
III	PE	Professional Elective-III®	4
III	PR	Project Stage I	12
		Total	20

SEMESTER - IV

Semester	Course Type	Course Name / Course Title	Total Credits
IV	PE	Professional Elective-IV®	4
IV	PR	Internship	4
	PR	Project Stage II	12
		Total	20

Professional Elective Tracks

Semester	Name of the Course	Type
II	Advances in Energy Engineering and Management	Program Elective - I
II	Polymer Industrial Manufacturing	Program Elective - I
II	Advanced Water and Wastewater Treatment	Program Elective - I
III	Principles of Energy Engineering	Program Elective - II
III	Process Data Analytics	Program Elective - II
III	Green Technology	Program Elective - II
III	Process Intensification	Program Elective - II
III	Polymer Waste Management	Program Elective - II
III	Energy Engineering and Technology	Program Elective - III
III	Process Safety Management	Program Elective - III
III	Catalyst Technology	Program Elective - III
III	Piping Design and Engineering	Program Elective - III
III	Refinery Systems	Program Elective - III
IV	Multiphase Reactor Design	Program Elective - IV
IV	Resources, Economics and Policies in Energy Engineering	Program Elective - IV
IV	Chemical Process Engineering	Program Elective - IV
IV	AI ML in Chemical Engineering	Program Elective - IV
IV	Bioprocesses and bio-chemicals	Program Elective - IV

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.